Enterprise Asset Management (EAM)

This case study explains an integrated enterprise asset management process in detail and thus fosters a thorough understanding of each process step and underlying SAP functionality.

Product

S/4HANA 2023
Global Bike

Fiori 3.0

Level

Beginner

Focus

Enterprise Asset Management

Authors

Robert Häusler
Chris Bernhardt
Stefan Weidner

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4.3

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MOTIVATION

Based on continuous automation of production processes, availability and quality of manufacturing facilities mens that plant maintenance becomes more important. Plant maintenance comprises the three following activities: inspection, maintenance and repair.

The focus is not only on availability of manufacturing facilities but rather further aspects like system safety, product quality, procedure optimization and environment protection play a decisive role.

SAP offers a flexible facility management and maintenance system to strengthen these factors and increase the success of a company.

PREREQUISITIES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to successfully work through this case study, it is not necessary to have finished the EAM exercises. However, it is recommended.

NOTES

This case study uses the Global Bike (GB) data set, which has exclusively been created for SAP UA global curricula.



SAP University Alliances

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Process Overview

Learning Objective Understand and perform an integrated maintenance process. **Time** 60 min

Scenario It is summer and the production line is getting into top gear for the upcoming event "Race Across America". There has been a sudden and unpredictable malfunction in a circulation fan in the air conditioning system. In order to process a complete maintenance process, you will take on different roles within the Global Bike company to solve the problem.

Employees involved

Jun Lee (Production Manager)
Jermain Kumins (maintenance technician)
Pamela Ross (Shop Floor Worker)

A fan in the production facility is defective. As a Shop Floor Worker, it is your task to create the malfunction report. Before you create the notification, you have to locate the exact defective part and record the technical details for the repair. After that, you have to plan the production operations until the final repair of the faulty fan unit. The defective parts shall be exchanged and followed by a function test. After the successful installation of the spare filter, you can finish the order technically. As a final step, you can check if all the maintenance tasks have been carried out correctly.

The following graphical illustration depicts the whole process (8 tasks).

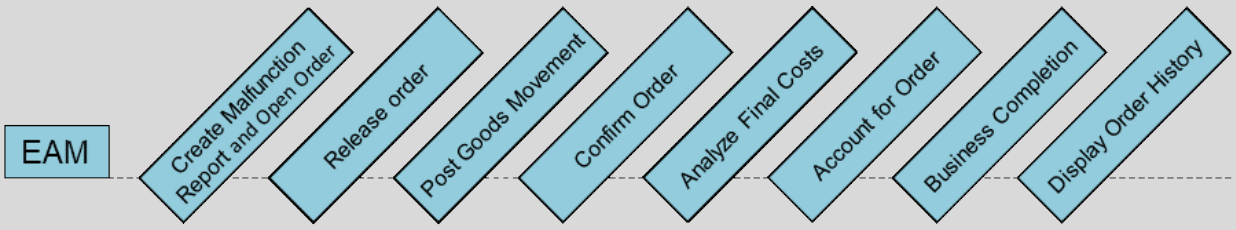


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Step 1: Create Malfunction Report and Open Maintenance Order

Task Create a malfunction report and open Maintenance Order from Malfunction Report

Time 10 min

Description Use the SAP Fiori Launchpad to create a malfunction report, propose a solution, and generate an order directly from the malfunction report.

Name (position) Pamela Ross (Shop Floor Worker)

Technical objects:

In the SAP system, technical objects are managed in the master data as Equipment and Functional Locations.

Equipment

Equipment refers to an individual, physical item that requires maintenance on its own (e.g., pumps, compressors, transformers, generators, batteries, valves, sensors). Equipment is typically assigned to a Functional Location, which is also immobile in nature (e.g., power plants, roads, piping systems).

Functional Location

Functional Locations are hierarchically organized structures that represent a technical system, a building, or a part of it. They can be structured spatially, functionally, and process-oriented.
When creating a Functional Location, it often assumes the role of the place where individual objects (motors, gearboxes, pumps, etc.) can be installed. In such cases, both the installation and removal locations can be seen from the perspective of the installation site as well as from the perspective of the individual installation or removal object.

Malfunction Report:

Malfunction Report

A **malfunction report** describes a fault in an object that restricts its performance in some way. The malfunction report is used to inform the maintenance department, for example, that a plant is not functioning correctly, its performance is reduced or completely fails, or poor results are being produced.

Start

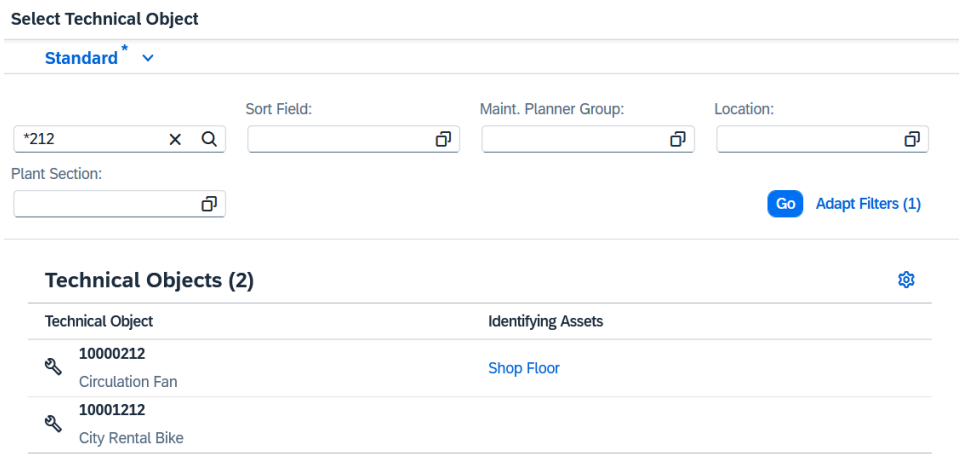
Go to the space *Enterprise Asset Management* in the role of *Shop Floor Worker* use the app *Manage Malfunction Report* to create a malfunction report and generate an order directly from the malfunction report.



Click on Create, to create a malfunction report.

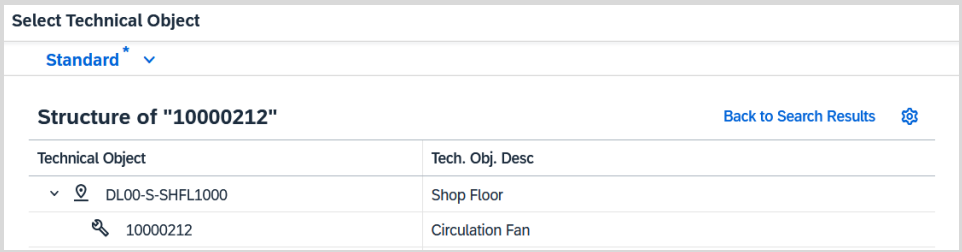
To select the *technical object (Equipment)*, use the . Enter *### in the search field (replace ### with your number, e.g., 212) and click on Go.

*###



Note As you can see, your equipment is located in the *functional location* of the **Shop Floor**. Click on *Shop Floor* to view it.

Shop Floor



The functional location is an organizational unit of logistics that structures the objects of a company to be maintained according to functional (e.g., drive unit), process-oriented (e.g., condensation), or spatial (e.g., hall) aspects. A functional location represents the location where a maintenance measure is to be performed.

Please double-click on your *Equipment* (**Circulation Fan**) to choose this.

Circulation Fan

Enter **Circulation fan defective ###** as the *description*. In the long text, enter **Circulation fan does not work correctly, Exchange filter**.

Circulation fan defective ###

Circulation fan does not work correctly, Exchange filter

Select **Production Restricted (2)** as the *Effect* using the button .

Production Restricted (2)



Click on Save.

Note In a real situation, the reporting person would not suggest a solution, as this is the responsibility of the relevant technical staff. However, for simplicity, a malfunction report with a subsequent solution suggestion has been created by the reporting person in order to directly generate a work order from the malfunction report.

Switch to the *Job Details* tab → *Work*. Click on Edit. In the *Edit Work Item* window, make sure that *Work Center/Plant* **MANT1000/DL00** is specified.

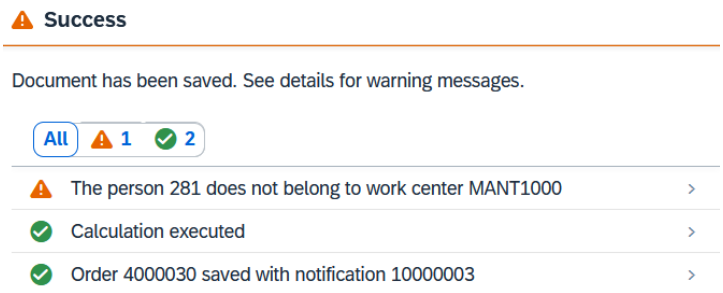
MANT1000 / DL00

Now replace the original *description* Circulation fan defective ### with **Exchange filter immediately**. Enter **2** as the *planned effort*, with the *unit* **hour**.

Exchange filter immediately
2
hour



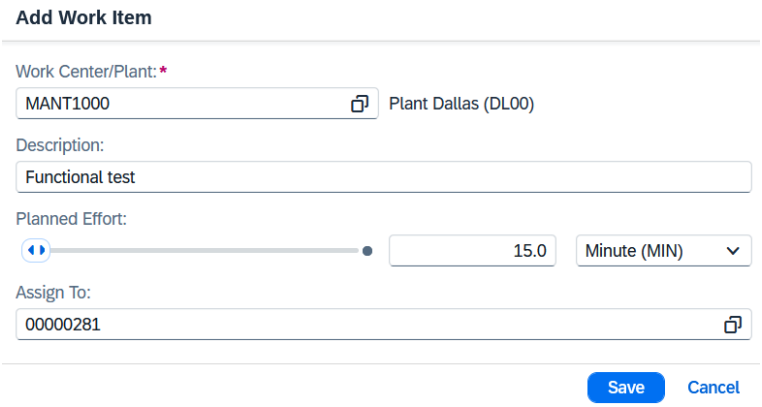
Click on Save. A warning message will appear. Click on *Show details*.



Click on OK. Create another work by opening a new line in the Job *Details tab (Work (1))*. Click on Add.

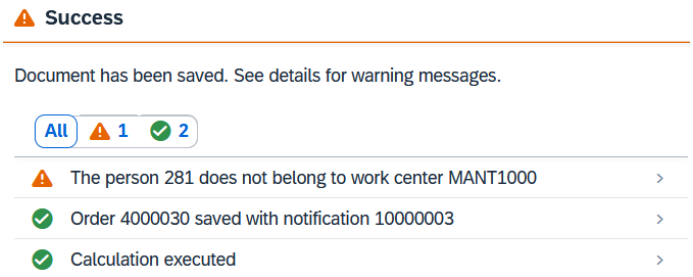
Select *Work Center/Plant* as **MANT1000/DL00**. Add **Functional test** as the *description* for the work and plan a duration of **15 minutes** for the work. Select *Assign to* **LEARN-###** (Personnel number).

MANT1000/DL00
Functional test
15
Minute
Learn-###



Click on Save to save the order. Click on *Show details*. Note down your order number.

Order number
.....

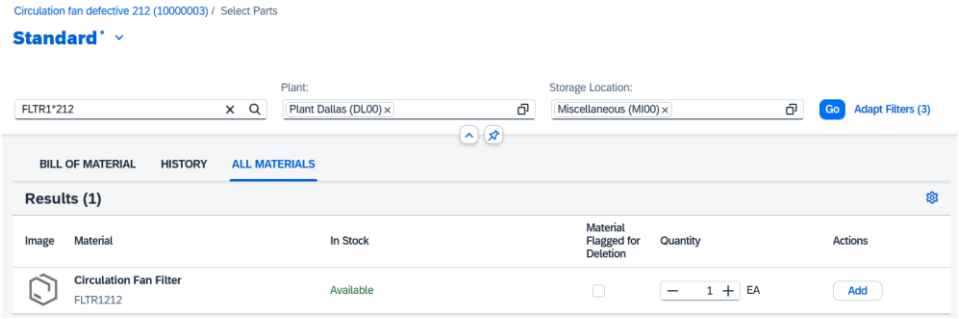


Click on OK.

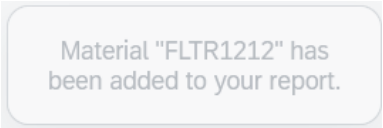
Note Work effort relates to the work that must be done. The number must be related to the employees. The duration defines the time when the object cannot be used for work. So, if the work effort is 8hrs and 2 people are working on it, work will be done in 4hrs. Another example is the varnishing of an object. Here the effort is lower than the duration since the object needs to dry some time and no work is done at the object within the drying process.

In the *Parts* tab, click on Add. Select the *All Materials* tab, then enter **FLTR1*###** (replace ### with your number) in the *search field*. Choose **DL00** as the *plant* and select **Miscellaneous (MI00)** as the *storage location* using the F4 help. Then, press Go. Select **1** as the *quantity*.

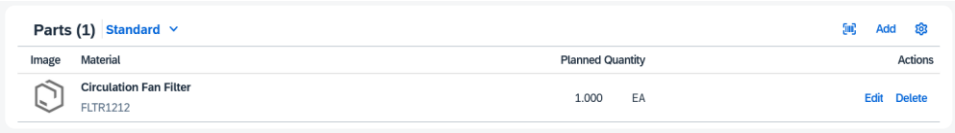
FLTR1###
DL00
MI00
1



And click on Add.



Click on ‹ to return to the *Process malfunction report*.



Click on the home icon to return to the Fiori Launchpad overview.

Step 2: Release Order

Task Release the Order. **Time** 5 min

Description Use the SAP Fiori Launchpad to release the order.

Name (Position) Jun Lee (Production Manager)

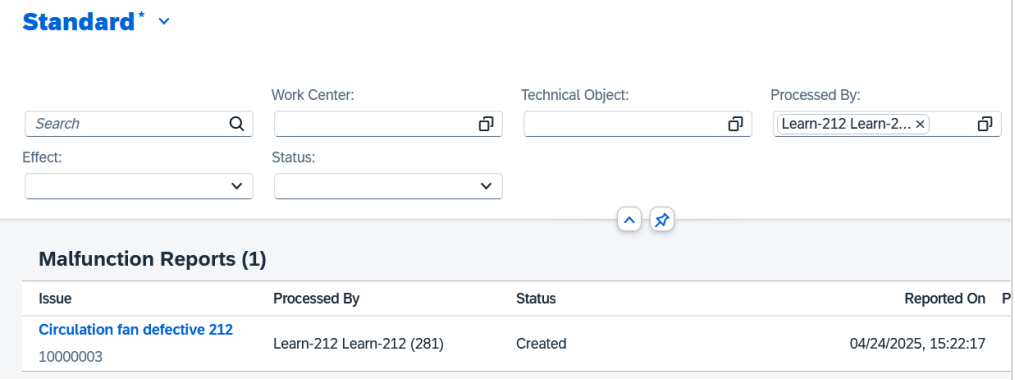
Go to the space *Enterprise Asset Management* in the role of *Production Manager* use the app *Manage Malfunction Reports*, to release the order.

Start

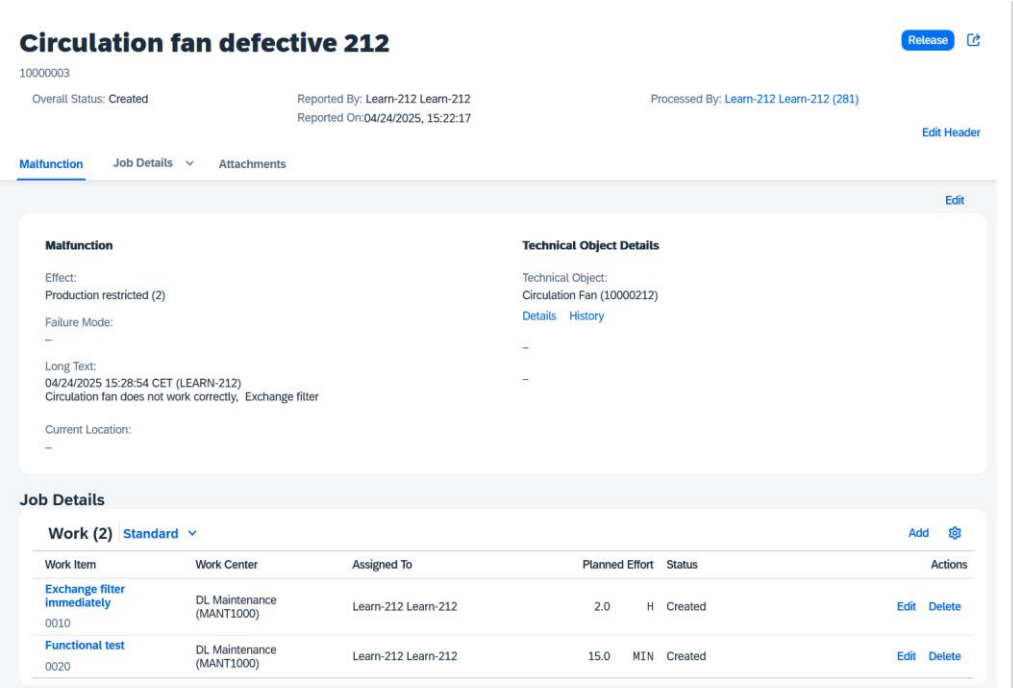


To open *Malfunction Report*, enter your **LEARN-###** in the *Processed By* field and click on Go.

LEARN-###



Press › to access the created *order*.



Select Release to release the order.

Note The order is released **after** saving it.When you release a maintenance order, the system checks the availability of materials and production resources or tools. At the time of release (at the latest), material reservations become relevant to materials planning, materials withdrawn, and purchase requisitions are generated.

You can only perform activities like print shop papers, withdraw material or book goods receipt after you released the order.

Click on the home icon to return to the Fiori Launchpad overview.

Step 3: Post Goods Movement

Task Post goods issue for the order. **Time** 5 min

Short Description Use the SAP Fiori Launchpad to check out the recirculation fan filter for the maintenance order.

Name (Position) Jermain Kumins (Maintenance Worker)

To book a goods issue, use the *Post Goods Movement* app in the space *Enterprise Asset Management* in the role of *Maintenance Worker*.

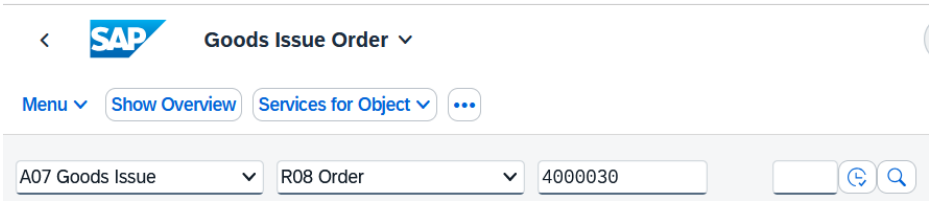
Start



In the *Goods Movement* screen, change the Material Document drop down to **Goods Issue**, adjust the type of Goods Issue to a **Order**. Enter the **order number** in the empty field next to it.

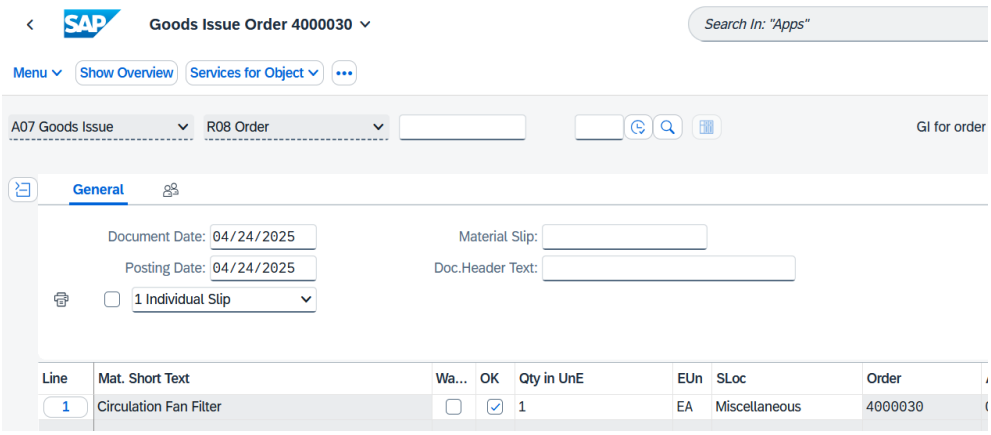
Goods Issue Order

Order Number



Press *Enter*. If it is displayed, select the **OK Checkbox**. Confirm your entries by pressing *Enter*. If this is not possible, minimize the lower detail data view by clicking on the icon .

OK



Material Document Number

Then, click on Post to save your receipt. The system will assign a unique material document number.



Click on SAP to return to the SAP Fiori Launchpad.

Step 4: Confirm Order

Task Confirm the order. **Time** 10 min

Description Use the SAP Fiori Launchpad to confirm the order.

Name (Position) Jermain Kumins (Maintenance Worker)

The overall completion confirmation is used to capture further data besides working times. For instance, used materials, information about errors, executed work and so on. All of this data is important for future maintenance history and analysis.

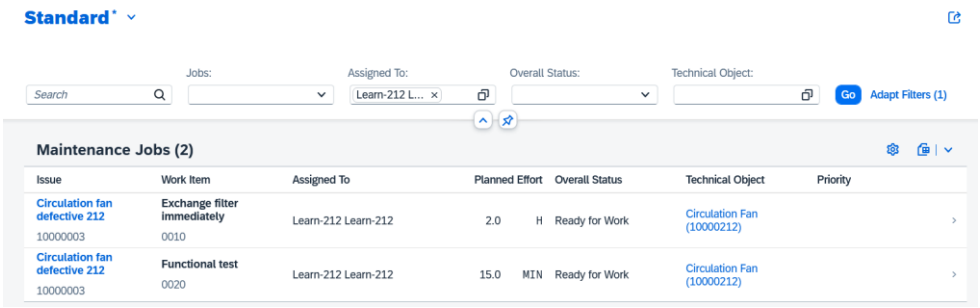
Go to the space *Enterprise Asset Management* in the role of *Maintenance Worker* use the app *Repair Malfunctions – My Job List* to confirm the order.

Start



Enter your **LEARN-###** in the *Assigned to* field and click on the button Go.

LEARN-###



Press the button › in the first line to access the created malfunction. There, select the *Job Details* tab.

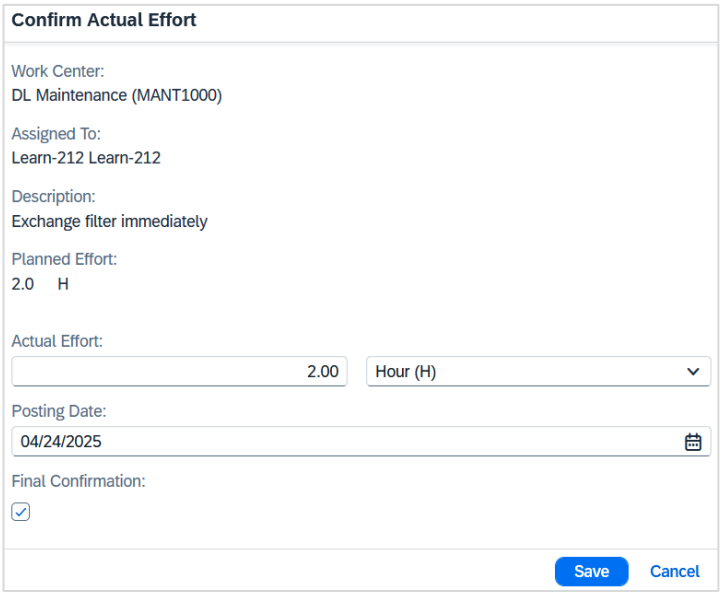
For the job *Exchange Filter Immediately*, change the *status* to **Work is done**. Afterwards, click on the Record Time button.

Work is done



2 hours
Final Confirmation

Make sure that **2 hours** are selected as the *actual effort*, and select the **Final Confirmation** option.



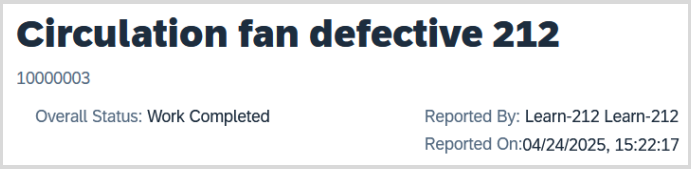
Select the button Save.

15 Minutes
Final Confirmation

Repeat the steps for the job *Functional Test*. Enter **15 minutes** as the actual effort and select the **Final Confirmation** option.

Back in the Order overview, click on the button at the top Complete. The order will be technically completed.

Note The two processes (and thus the order) have now been reported back. The overall status is: "Work Completed." This means that the work is finished, the target times were met, and a filter was removed from the warehouse.



Click on the home icon SAP to return to the Fiori Launchpad overview.

Step 5: Analyze Final Costs

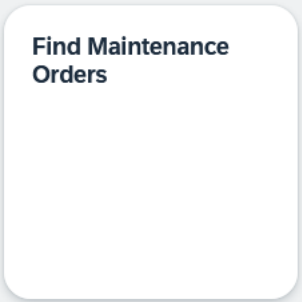
Task Analyze the final costs. **Time** 5 min

Description Use the SAP Fiori Launchpad to execute a cost analysis.

Name (Position) Jun Lee (Production Manager)

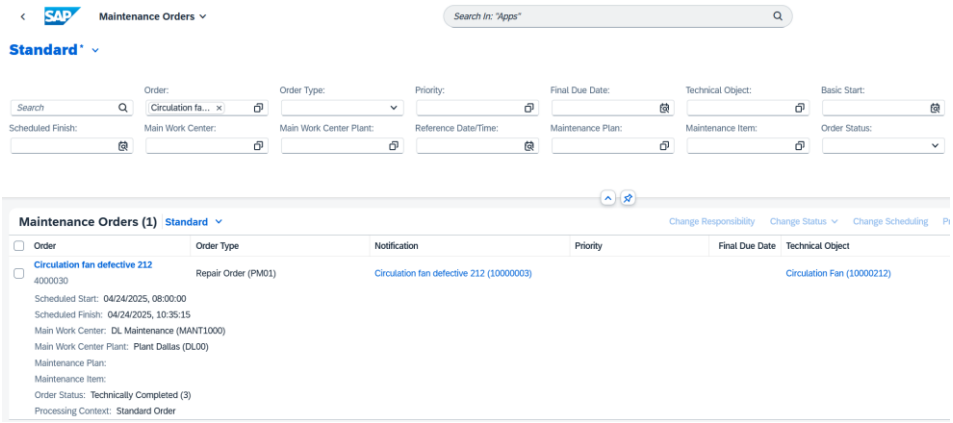
Go to the space *Enterprise Asset Management* in the role of *Production Manager* use the app *Find Maintenance Orders* to execute a cost analysis.

Start



Enter the **number of your most recently processed order** as the *order*. Click on the button Go.

Order Number



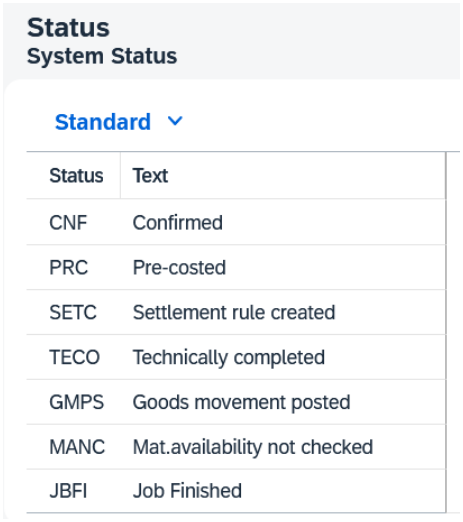
Click on the line of your order. Now, select the *Costs* tab to view the costs for the order. You will receive an overview of the current planned and actual costs.

Costs

Standard

Value Category	Estimated Cost		Planned Cost		Actual Cost	
Internal Labor	0.00	USD	0.00	USD	25.31	USD
External Services	0.00	USD	0.00	USD	0.00	USD
Stock Material	0.00	USD	200.00	USD	200.00	USD
Non-stock Material	0.00	USD	0.00	USD	0.00	USD
Overhead Rates	0.00	USD	76.00	USD	79.80	USD
Miscellaneous	0.00	USD	0.00	USD	0.00	USD
Total	0.00	USD	276.00	USD	305.11	USD

Now, select the *Status* tab to view the system status.



Status
System Status

Standard

Status	Text
CNF	Confirmed
PRC	Pre-costed
SETC	Settlement rule created
TECO	Technically completed
GMPS	Goods movement posted
MANC	Mat.availability not checked
JBFI	Job Finished

Click on the home icon to return to the Fiori Launchpad overview.

Step 6: Accounting for the Maintnenance Order

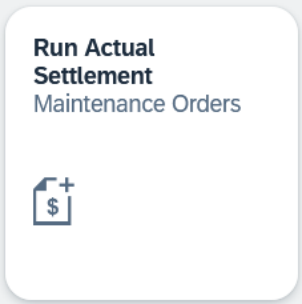
Task Accounting for the Maintenance Order. **Time** 5 min

Description Use the SAP Fiori Launchpad to settle the accrued costs with the settlement receiver.

Name (Position) Jun Lee (Production Manager)

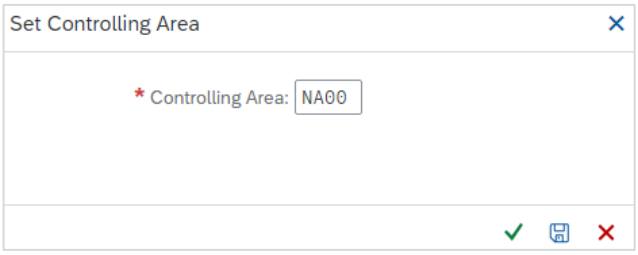
Go to the space *Enterprise Asset Management* in the role of *Production Manager* use the app *Run Actual Settlement* to account for the order.

Start



If the *Set Controlling Area* pop up opens enter **NA00** as Controlling Area.

NA00

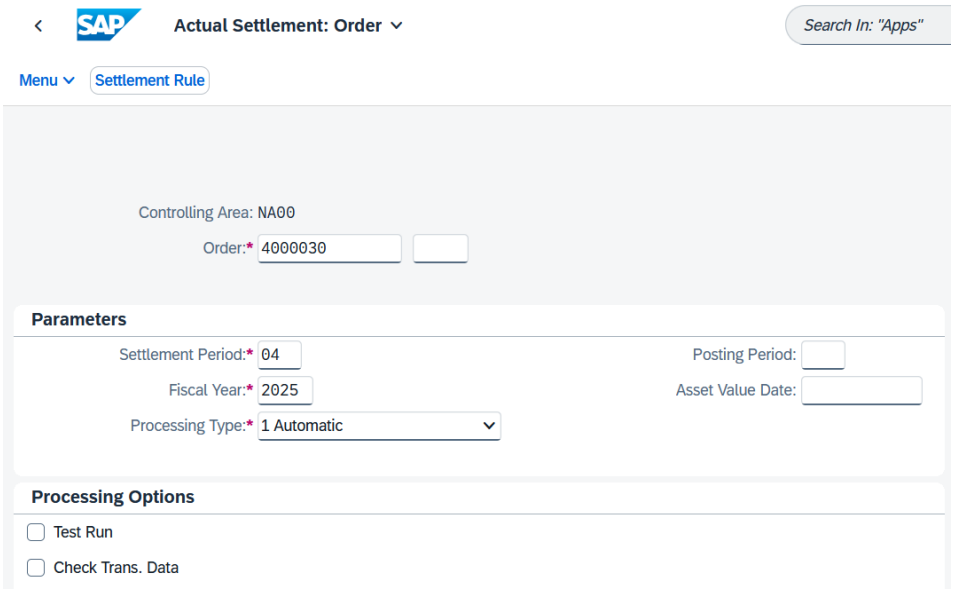


If your order number does not appear, either enter it yourself or search for it using F4. Additionally, make the following entries: Enter the **current month** as *Settlement period* and the **current year** as *Fiscal Year*. Make sure that the Processing type is set to **automatic**.

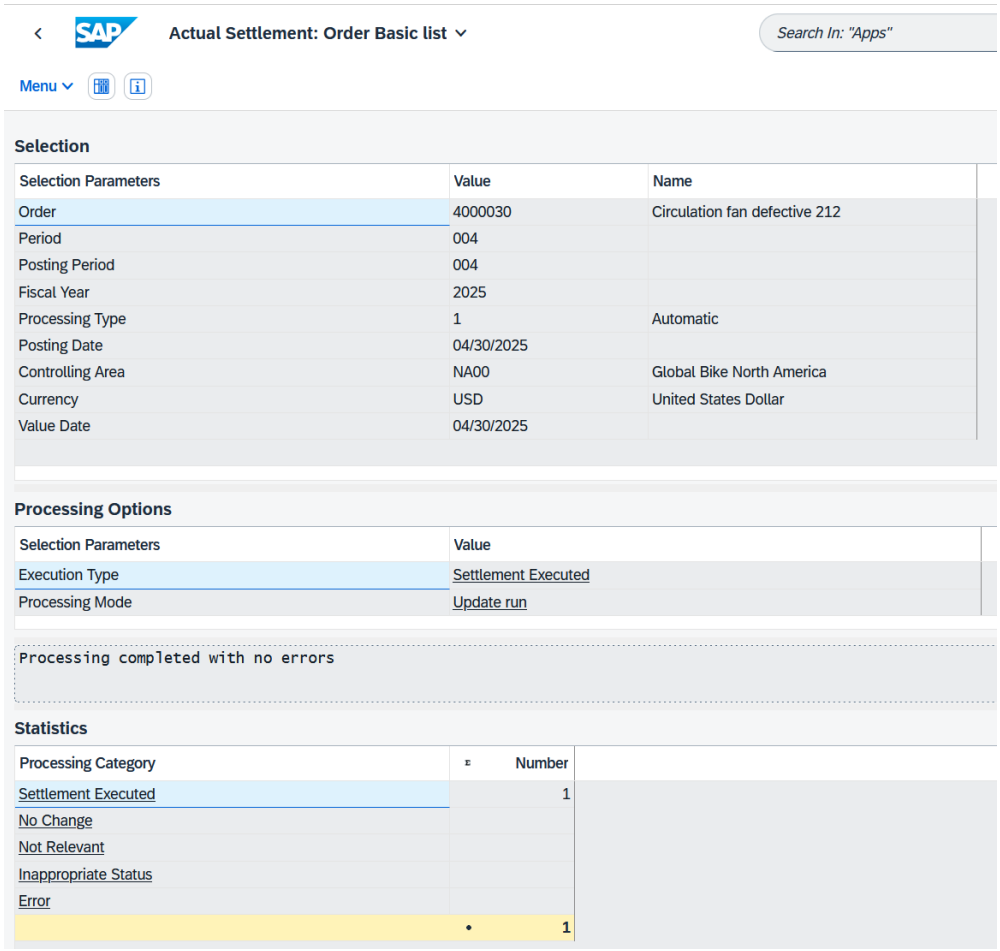
current month
current year
automatic

Remove the **Test Run** indicator. Compare with the next screenshot.

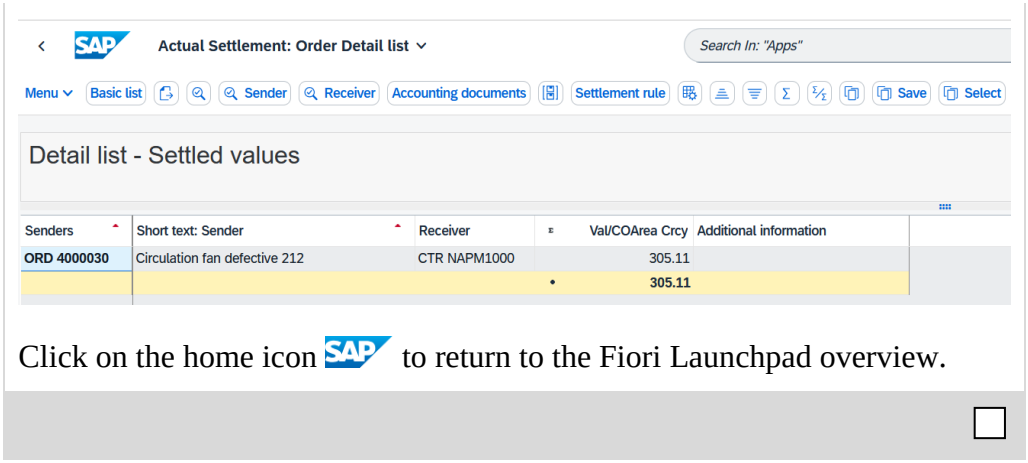
~~Test Run~~



Click on Execute.



Click to get to the detailed list of settlement.



Click on the home icon to return to the Fiori Launchpad overview.

Step 7: Compelete Order (Business)

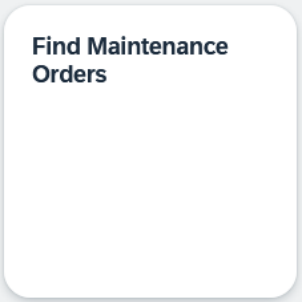
Tasks Analyze costs and complete the order. **Time** 10 min

Description Use the SAP Fiori Launchpad to finish the order.

Name (Position) Jun Lee (Production Manager)

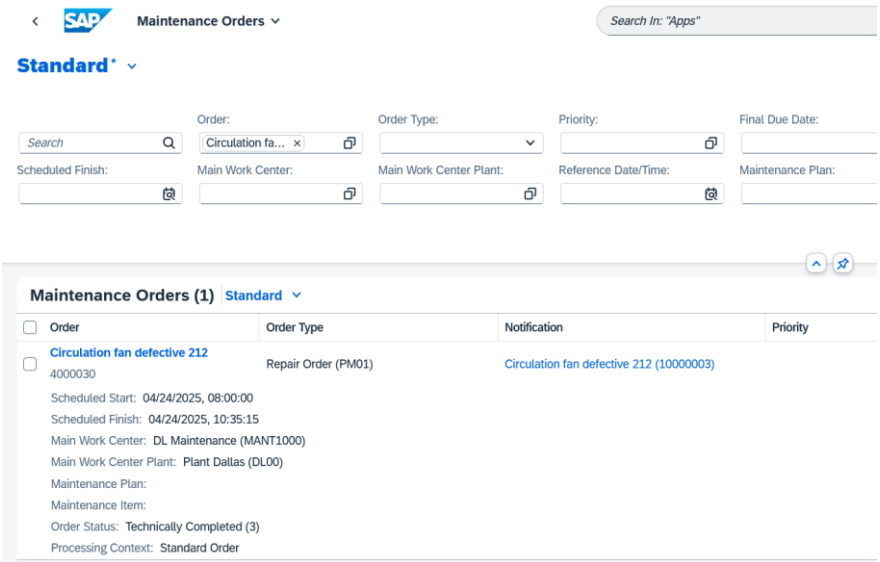
Go to the space *Enterprise Asset Management* in the role of *Production Manager* use the app *Find Maintenance Orders* to finish the order.

Start



Enter the **number of your most recently processed order** as *Order*. Click on the Go button.

Order Number

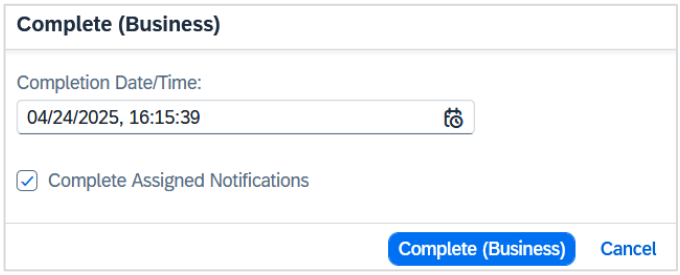


Select your order and click the button Change Status, then choose *Complete (Business)*.



current date and time
Complete assigned notifications

In the dialog window, you will see the **current date and time** as the *completion date/time*. Also, select **Complete assigned notifications.** Then, press the Complete (Business) button.



Note After the commercial completion, the order can no longer be charged with additional costs.

Click on the home icon to return to the Fiori Launchpad overview.

Step 8: Display Order History

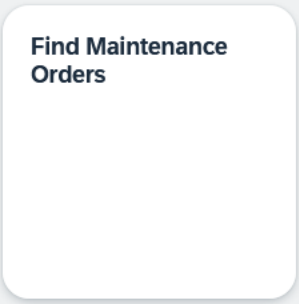
Task Display the order history for your equipment. **Time** 5 min

Description Use the SAP Fiori Launchpad to see a detailed history about the equipment.

Name (Position) Jun Lee (Production Manager)

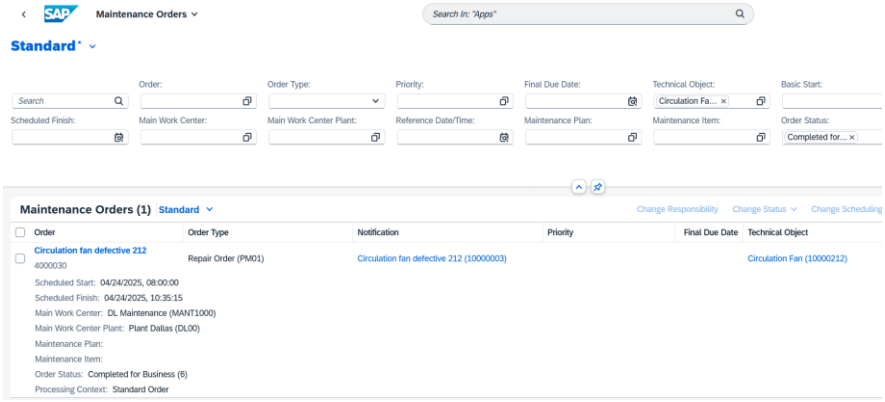
Go to the space *Enterprise Asset Management* in the role of *Production Manager* use the app *Find Maintenance Orders* to display the order history.

Start

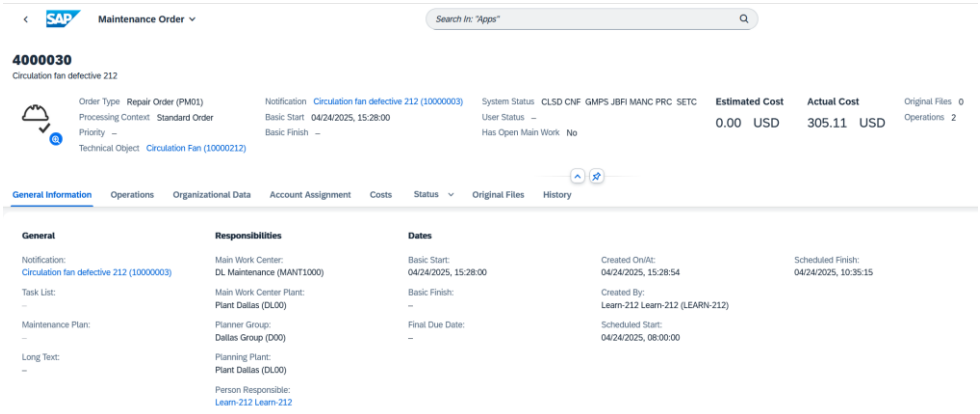


Select the *order status* **Completed for Business** enter the *technical object* as **10000###** (replace ### with your number). Click on the button Go.

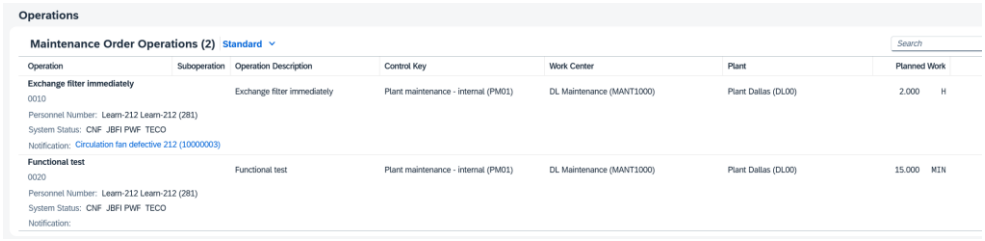
Completed for Business
10000###



Click on › the button to view a detailed history of the equipment.



Tab *Operations:*



Tab *Costs:*



Value Category	Estimated Cost		Planned Cost		Actual Cost	
Internal Labor	0.00	USD	0.00	USD	25.31	USD
External Services	0.00	USD	0.00	USD	0.00	USD
Stock Material	0.00	USD	200.00	USD	200.00	USD
Non-stock Material	0.00	USD	0.00	USD	0.00	USD
Overhead Rates	0.00	USD	76.00	USD	79.80	USD
Miscellaneous	0.00	USD	0.00	USD	0.00	USD
Total	0.00	USD	276.00	USD	305.11	USD

Click on the home icon to return to the Fiori Launchpad overview.

EAM Challenge

Learning Objective Understand and perform an integrated EAM process. **Time** 30 min

Motivation After you have successfully worked through the *Enterprise Asset Management* case study, you should be able to solve the following task on your own.

Scenario Another fan in the production facility DL00-S-SHFL1000 has failed. The cause is a crack in the rotor blade (1001) due to wear (1004), caused by natural wear and tear from extended operation over time. Please initiate the repair of the fan. Select equipment 10000### and assign the order to the main work center MANT1000. The reporting user is LEARN-###. The estimated repair duration is 1.50 hours, with an actual feedback of 2 hours. No materials are required for the repair. After completion, the order will be technically closed, settled, and commercially closed.

Task Information Since this task is based on the *Enterprise Asset Management* case study you can use it as guidance. However, it is recommended that you solve it without any help in order to test your acquired knowledge.

